

UCE309: BUILDING MATERIALS

L	T	P	Cr
2	0	2	3.0

Course Objective: To expose students to the various building and general construction products and their associated quality, durability and availability.

Cement: Manufacturing, raw ingredients, chemical and mineralogical composition and types of cement.

Aggregates: Classification of aggregates, application and advantages of Manufactured Sand.

Cement mortar and concrete: Classification, properties and applications, water cement ratio, fresh, mechanical and durability properties of concrete.

Production of Concrete: Batching, mixing, transportation, placing, compaction and curing of concrete, quality control of concrete, concrete mix design.

Special concrete: Introduction to fibre-reinforced concrete, self healing concrete, lightweight concrete.

Chemical admixtures: Accelerating admixture, water reducing admixture, retarding admixture, air-containing admixture.

Bricks and blocks: Classification and characteristics of bricks, AAC blocks, hollow concrete blocks.

Steel: Manufacturing of steel, market forms of steel, classification of reinforcing steel bars

Miscellaneous building materials: Timber, glass, plastic, water proofing materials, paints and varnishes.

Emerging civil engineering materials: Introduction to refractories, laminate, adhesives, graphene and carbon composites.

Laboratory Work

1. Tests on cement (fineness, specific gravity, consistency and strength test).
2. Tests on fine and coarse aggregates (gradation, specific gravity, water absorption).
3. Design of concrete mix for required grade of concrete.
4. Tests on fresh concrete (workability test).
5. Tests on hardened concrete (compressive strength, flexural strength and split tensile strength).

Experiential Learning (ELC) Activity

Casting of concrete canoe using innovative lightweight concrete.

Course Learning Objectives (CLO)

The students will be able to:

1. Design the mix proportions for required grade of concrete.
2. Assess the quality of procured materials such as bitumen, asphalt, insulating materials, glass, and water proofing materials, paints and varnishes on site.
3. Evaluate the quality of building materials such as timber and steel.
4. Perform experiments in laboratory to evaluate properties of construction materials such as cement, aggregate, mortar and concrete.

Text Books

1. Gambhir M. L., Concrete Technology, Tata McGraw Hill Publishing Co. Ltd., New Delhi, 2004.
2. Rangawala S. C., Engineering Materials, Charotar Publishing House, Anand, 1992.
3. Chaudhary S. C., Patel K. A., Testing & Evaluation of Civil Engineering Materials, All India Council for Technical Education (AICTE), 2023.

Reference Books

1. Shetty M. S., Concrete Technology: Theory and Practice, S. Chand & Company, 2010.
2. Kumar Sushil, Engineering Materials, Metropolitan Press, 1994.
3. Kumar Sushil, Building Construction, Standard Publishers and Distributors, 1990.
4. Punmia B. C., Building Construction, Laxmi Publishing House, 1993.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	35
3	Sessional: (May include the following) Assignment, Regular Lab assessment and Quizzes, Project (Including report, presentation etc.)	40